

WEST AFRICAN SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

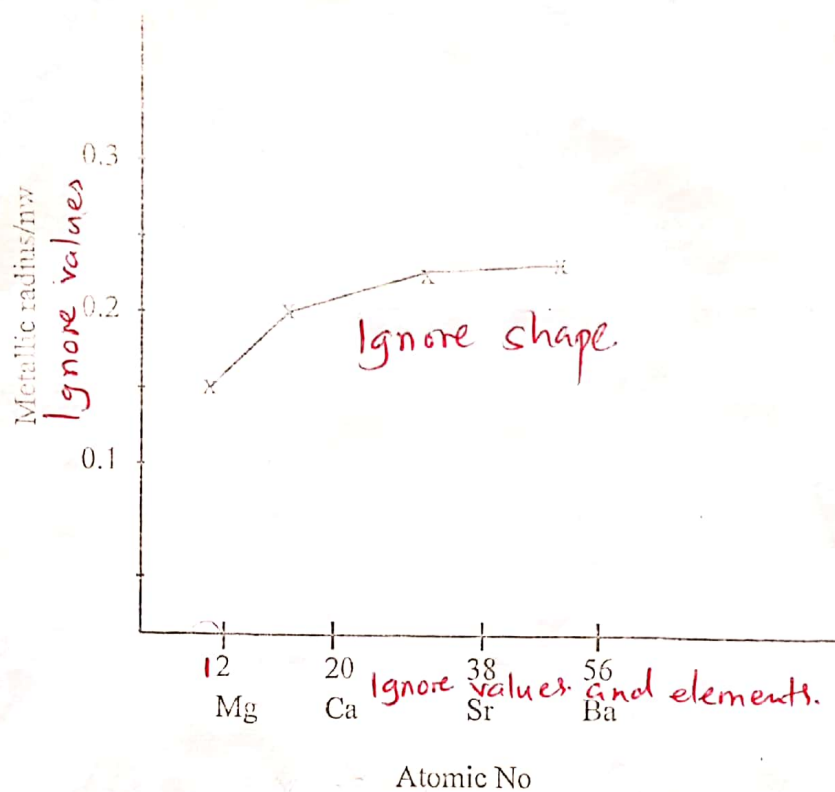
MARKING SCHEME

NOVEMBER 2021 SESSION

CHEMISTRY

6031/3

1s²2s²2p⁶3s²3p⁶ [1]
 accept electrons-in-boxes diagram that is
 correct and showing all orbitals.

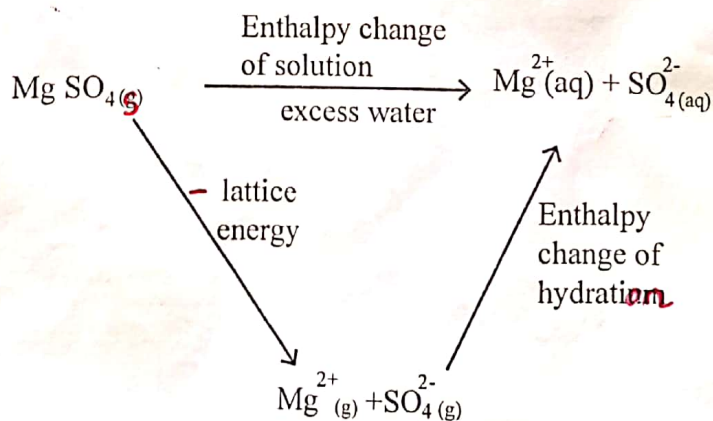


Labelled axes [1]
 Correct trend [1]

(iii) As metallic radius increases down the group, shielding effect increases as (larger distance of outermost electrons outweighs attraction of higher nuclear charge). [1]
 [1] as number of shells increase
 [1]
 [1]

~~It becomes easier for metallic atoms to form the +2 ions.~~

(b) (i)



accept energy level diagram

Correct formulae and state symbols

[1]

Correct direction of arrows

[1]

- (ii) Due to the large size of the SO_4^{2-} ion, there is a small difference between the lattice energies of its compounds. / AW

[1]

Down the group, the enthalpy change of hydration becomes less negative / as exothermic / decreasing solubility.

[1]

- (iii) Ba^{2+} ions are used to test for SO_4^{2-} in solution (a BaSO_4 is easily precipitated) / testing for SO_4^{2-}

[1] / less exothermic / more positive

[1]

- (c) (i) Used to produce plaster (of Paris) for immobilising broken limbs.

[1]

When water is added it expands slightly and sets quickly to form a hard solid / AW.

[1]

- (ii) Used for treatment of too acidic soils. / AW

[1]

It is basic.

[1]

- 2 (a) (i) Ionic lattice

[1]

- (ii) 1. Stronger ionic lattice in B than A / U3
Ionic compound of B divalent ions whereas in A it has monovalent ions / B has a higher charge density than A

[1]

2. D has a giant molecular lattice whereas E has a simple molecular lattice.

[1]

- (iii) 3. Ions are free to move; the oxide does not dissolve in water / 13.
different

[1]

- (b) (i) (Catenation) formation of many organic molecules/polymers

[1]

- (ii) Bond energy kJ/mol
C - C (350) high c-c bond energy values
~~C - O 360~~

[1]

~~Carbon can share up to four electrons with another atom;~~
~~Forming strong stable single/double/triple bond;~~ molecules

[1]

[1]

~~C - C bond has approximately same bond enthalpy as C - O bond.~~
~~Compounds with C - C bond stable~~

[1]



PbO reacts with dilute HNO_3 to produce a salt but PbO_2 does not react [1]
 $\frac{56}{100} \times 180 = 100.8 \text{ g}$ [1]

2. Theoretical mass of $\text{Pb}(\text{NO}_3)_2$ [1]

$$\frac{100.8}{203} \times 223$$

$$94.05 \text{ g}$$

[1]

% yield

$$\frac{72 \text{ g}}{94.05} \times 100$$

ecf

$$77.6\%$$

[1]

(iv) SO_2 causes chest pains/forms acid rain/destruction of vegetation/corrosion of buildings/lowering of soil pH through acid rain. [1]

3 (a) In any equilibrium, when a change is made to some external factor (such as temperature or pressure), the change in the position of equilibrium is such as to tend to change the external factor in the opposite direction/AW. [1]

(b) (i) $4.22 \times 10^6 - (7.27 \times 10^6 - 3.41 \times 10^6)$

$$= 0.36 \times 10^6 / 3.6 \times 10^5 \text{ Pa. [1]}$$

$$(ii) K_p = \frac{P_{AB}^2}{P_{A_2} \times P_B}$$

$$= \frac{(7.72 \times 10^6 \text{ Pa})^2}{(3.41 \times 10^6 \text{ Pa}) \times (0.36 \times 10^6 \text{ Pa})}$$

substitution [1]

$$= 48.55$$

ecf

[1]

(c) (i) 1. The forward reaction is exothermic hence the production of ammonia is favoured by low temperatures [1]

2. Low temperature lower the speed of reaction.

$\therefore$ A compromise temperature is used. [1]

(Iron) Catalyst is used to reach equilibrium faster. [1]

3. High pressures favour the forward reaction but this also increases operating costs/and the chances of explosions.

$\therefore$ Lower pressures are therefore used /AN [1]

Low/compromise

1 mark for all 3 correct conditions

Eutrophication [1]

[13]

Increase in BOD in water bodies leading to death of aquatic species / AW - clogging of water ways accept blue-baby syndrome [13]

1. ~~Nutrients promote the growth of water plants which are washed into rivers and lakes~~ [1]
2. ~~When the algae die, their decomposition uses up all the oxygen in water so that it cannot support any form of life.~~ [1]

[13]

Decrease in solubility of a sparingly soluble ionic compound when mixed with a solution containing one of its ions / AW [13]

- (i) ~~It is an ion that is found in both the solution of a compound and the sparingly soluble salt solution / AW~~ [1]
- (ii) It contains a common ion, of sodium Na^+ . Addition of the Na^+ ions reduces the solubility of the sodium stearate because its solubility product will have been exceeded. [13]



[1]

$$K_{sp} = [\text{C}_{17}\text{H}_{35}\text{COO}^{-}] [\text{Na}^{+}] \text{ for equation}$$

[1]

$$\text{Since at equilibrium } [\text{C}_{17}\text{H}_{35}\text{COO}^{-}] = [\text{Na}^{+}]$$

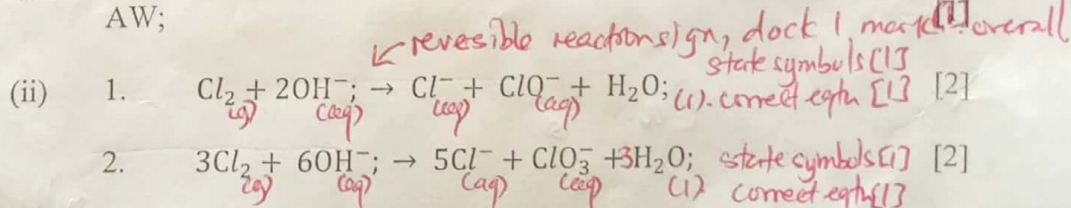
$$K_{sp} = [5 \times 10^{-4} \text{ mol dm}^{-3}]^2 \text{ accept correct substitution}$$

[1]

$$= 2.5 \times 10^{-7} \text{ mol}^2 \text{ dm}^{-6}$$

Reject without units.

- 4 (a) (i) A r in which a species is reduced and oxidized simultaneously / AW;

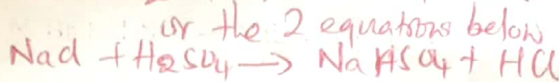


- (b) (i) reactivity decreases down the group; atomic size increases hence the power to attract an electron during reaction decreases / effective nuclear charge of halogen decreases HF [13] Highest bond energy [13] [1]
- (ii) Strong hydrogen bonding between HF and H_2O molecules requires a large amount of energy to break. [1]

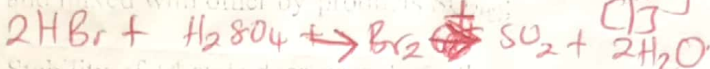
- (c) (i) inter electron pair repulsion between bonded F atoms is high; pushing nuclei apart / shorter bond length in F-F stronger repulsion [1]

- (ii) atomic radius increase down the group; [13] ability of the nucleus to pull electron towards itself becomes low / [1]

Shielding effect is increasing down the group effective nuclear charge decreases



HBr produced is further oxidised to Br_2 (alternatively the eqn below)



out,
The electron in Br^- is more
screened than the 1 in
 Cl^- ion / An [13]

Inert pair effect more pronounced down the group [1]

e.g. CO_2 and SiO_2 are stable;

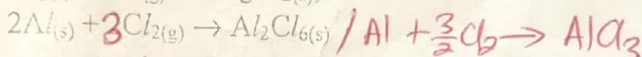
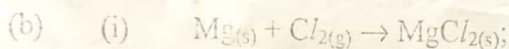
unlike CO and Si which are in the +2;

E° values become more positive down the group;

s-pair of electrons is less affected by screening [13]

(ii) CO_2 is more stable; CO is unstable; [13]

CO reacts exothermically to form CO_2 ; [13]



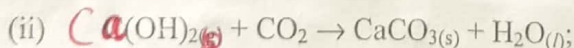
Ignore state symbols

Al_2Cl_6 reacts more vigorously with water than MgCl_2 . [13]

(ii) MgCl_2 a giant ionic structure;
dissolves in water; to form a neutral solution;
 Al_2Cl_6 dissolves in water releasing ions;
has a simple molecular structure forming an acidic solution; [1]

(iii) MgCl_2 neutral; Al_2Cl_6 forms a more acidic solution than that of MgCl_2 ; [1]
 Al_2Cl_6 acidic Al^{3+} ions polarize water molecules to a greater extent [13]. [1]

(c) (i) decreases down the group [13]
cationic radius increases, decreasing charge density;
polarizing power of M^{2+} cation decreases CO_3^{2-} suffers less distortion down the group; [1]



(visible) white precipitate (of CaCO_3) formed; / milky/chalk. [1]



Excess CO_2 reacts with $\text{CaCO}_3(\text{g})$ (white ppt) forming $\text{Ca(HCO}_3)_2$
A clear colourless/ milky/ness disappears; [1]

6 (a) (i) no reaction with Cl_2/Br_2 in the dark; [1]

accept any electrophilic addition reactn.

no reaction with steam/HX or KMnO_4 ; [1]

no reaction no oxidation reaction with KMnO_4 .

(ii) delocalization of π -electrons; [1]

ring of electrons

5

an acid

has an acidic phenolic group.

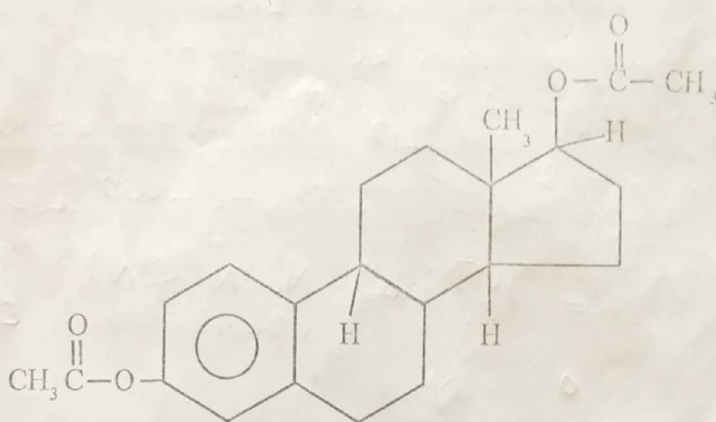
~~hydroxyl~~

mentone brings alcohol and phenolic group

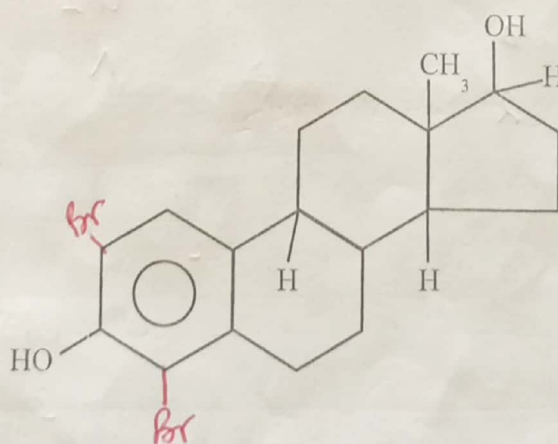
(ii) 1. (Phenol and OH groups present); on form hydrogen bonds with water;

2. hydrogen bonding (and van der Waals forces) require higher energy to separate

(iii) 1.



2.



(iii)

(iv) 1.

White fumes of HCl / sweet smell

2.

white precipitate

decolourisation of Br₂;

alternatively - white fumes } any 2.
white fumes

find alternatives to CFCs / reducing use of CFCs [1] FHC

(ii) structural isomerism (positional) [1]

(iii) A does not react C-Cl is a partial double bond / OH repelled by ring electrons [1]
B reacts/undergoes nucleophilic substitution/reaction equation [1]

methyl secondary alcohol

Test 1 (b) $\text{CH}_3-\text{CH}(\text{OH})-\text{R} / \text{CH}_3-\text{C}(=\text{O})-\text{R}$ [1] (b) methyl ketone for each (b) salvage mark if both functional (b) groups without showing deduction

Test 2 absence of methyl ketone / $\text{CH}_3-\text{C}(=\text{O})-\text{R}$ not present [1]

Test 3 presence of (carboxyl) $-\text{COOH}$ [1]

Therefore functional groups are $\text{CH}_3-\text{C}(=\text{O})-\text{R}$ and $-\text{C}(=\text{O})-\text{OH}$ [1]

(c) $(\text{CH}_2\text{O})_n = 90,08$ molecular formula = $\text{C}_3\text{H}_6\text{O}_3$ [1]
 $\Rightarrow 30n = 90,08$ structural formula = $\text{CH}_3-\text{CH}(\text{OH})-\text{COOH}$ [1]

(i) 1. nucleophilic substitution [1]

2. elimination [1]

(ii) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ [1] (iii) (NaCN/KCN) ethanol, reflux [1]

8 (a) (i) A $\text{CH}_3\text{COCH}_2\text{CH}_2\text{COOH}$; [1]

accept displayed formula

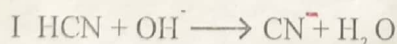
Y/A CH_3OH ; [1]

(ii) 1 Type of reaction is (acid) hydrolysis; [1]

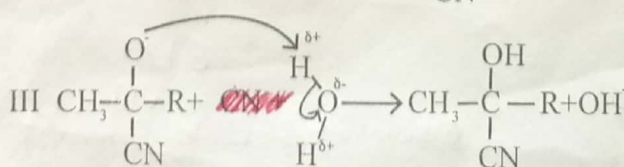
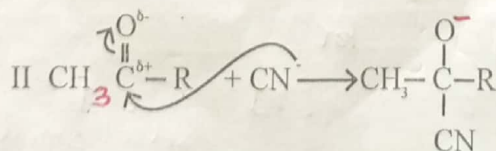
(iii) B $\text{CH}_3\text{C}(\text{OH})(\text{CH}_2\text{CN})\text{COOCH}_3$; [1]

(iv) 2 Reaction II is nucleophilic addition; [1]

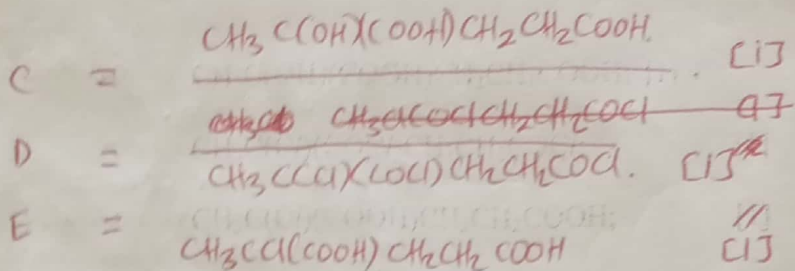
(b)



Taking R = $\text{CH}_2\text{CH}_2\text{COOCH}_3$



accept displayed formulae.



d) D/E is acidic [1]
presence of carboxyl group [1]

1c)

(iv) 1. pH of resulting solution is between 1-2/acidic; [1]
2. This is due to formation of HCl which is a strong acid; [1]
(ii) (acidic) [1]
It is a mixture of acids; Are corrosive and affect the environment; [1]
Causes irritation to area of contact [1]
They react with biomolecules/proteins on the human body; [1]

9 (a) (i) I Fe_2O_3 ; [1]

II $3\text{H}^+ / +3$; R without polarity [1]

(ii) either $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ /
 ~~$2\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Fe} + 3\text{CO}_2$~~ [1]

(b) I $\text{Fe}(\text{OH})_3 + e^- \rightarrow \text{Fe}(\text{OH})_2 + \text{OH}^-$ $E^\theta = -0.55 \text{ V}$ [1]
 ~~$\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$ $E^\theta = +0.77 \text{ V}$~~
 E^θ is negative showing that reaction is less likely to occur, hence Fe^{3+} is more stable than Fe^{2+} in sodium hydroxide; [1]

II $\text{Fe}(\text{CN})_6^{3-} + e^- \rightarrow \text{Fe}(\text{CN})_6^{4-}$ $E^\theta = +0.36 \text{ V}$ [1]
 E^θ is positive showing that $\text{Fe}(\text{CN})_6^{4-}$ is more stable than $\text{Fe}(\text{CN})_6^{3-}$; [1]

(c) (i) - method of separation of charged particles. [1]
- Digested sample in a solution of either a base or an acid; [1]
- Amino acid exist as cation in an acid or anion in a base; [1]
- Column should be charged opposite to the charge on the amino acids; [1]
- separation is based on electrostatic forces of attraction between the resin charge amino acids/particles [1]
- ~~ions are driven by a solution of a simple salt down the column~~ [1]
(ii) pH gives amino acids a charge so that they can interact with the charged resin/column during separation; [1]

(d) (i) - CO_2 produced is a greenhouse gas which leads to global warming; / Non-renewable [1]
- combustion in engines produces NOX which can catalyse the formation of acid rain; (any one) [1]
- SO_2 causes acid rain / produces CO which is toxic [1]
(ii) Suggest two alternatives sources of energy; [2]

Column is filled with a resin containing fixed charges [1]
Mixture to be separated is added to the column [1]
Ions are driven by a solution of a simple salt down the column [1]
Ions sep [1]

- Solar power / Sunlight
- Wind
- Hydroelectric power
- Fuel cells (oxygen-hydrogen)
- Nuclear power
- Geothermal energy
- Hydrogen gas
- Tidal energy

Materials made up of particles with dimensions from 1 to 100 nm.

[13]

(They are more chemically reactive and can easily form harmful free radicals with oxygen)

[13]

They can be readily taken up by the body through the skin [13]
They can gain access to the bloodstream following inhalation [13]

They can cross biological membranes and nuclei [13]

[maximum 2 marks] [any 2]

[13]

(b) (i)

- Poisoning of fish and other marine life [13]

- can be washed back ashore contaminating water supplies [13]

- Eutrophication

(ii)

- heavy metals and other toxic substances can leach out and contaminate nearby water supplies; [13]

- anaerobic conditions can lead to production of toxic H_2S and greenhouse gas methane; [13]

- chlorine containing material can form toxic dioxin if temperatures are not controlled; [13]

- Takes up land that can be used for agricultural/construction purposes containing fixed charges [13]

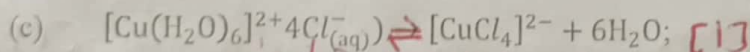
(iii)

Waste water flows over resin beds.

Metal ions are adsorbed on the surface of the beds and H^+ ions are released back into the water; [13]

Anions are adsorbed by anion exchanges and replaced with OH^- ions; [13]

H^+ ions combine with OH^- ions to form water; [13]



reject in one way arrow.

When concentrated hydrochloric acid is added.

The equilibrium shifts to the right;

ligand exchange [13]

A green solution is produced;

due to the blue Cu^{2+} and yellow $[CuCl_4]^{2-}$; [13]

Eventually colour turns yellow;

due to formation of more $[CuCl_4]^{2-}$; [13]

- soil toxicity

- production of leadate

- outbreak of disease

- makes place an eyesore

- Outbreak of rodents/vectors [13]

[any 3 marks]